

Lecture 6: Balanced Search Trees

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September 17, 2026

601.433/633 Introduction to Algorithms

Announcements

- ▶ HW1 due now, HW2 released
 - ▶ Oral presentations!
 - ▶ Form group of 3 (required!) in Canvas, sign up for presentation slot as a group in Canvas.
 - ▶ Format: HW has three questions. Grader will randomly assign each student to present one question. Rest of group can chime in occasionally.
- ▶ Regrade policy: 120 hours (five days) from when grades released
 - ▶ Don't abuse this!
 - ▶ If too many of your regrade requests do not result in positive changes, will ban you from regrade requests
 - ▶ Grading can go down!

Introduction

Today, and next few weeks: data structures.

- ▶ Since “Data Structures” a prereq, focus on advanced structures and on interesting analysis

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Today and later: data structures for *dictionaries*

Definition

A *dictionary data structure* is a data structure supporting the following operations:

- ▶ **insert(key,object)**: insert the (key, object) pair.
- ▶ **lookup(key)**: return the associated object
- ▶ **delete(key)**: remove the key and its object from the data structure. We may or may not care about this operation.

Obvious Approaches

Reminder: all running times for *worst case*

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Approach 1: Sorted array

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Goal: $O(\log n)$ for both.

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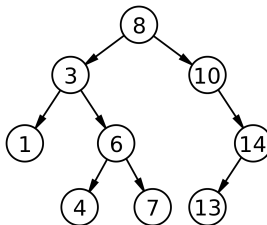
Goal: $O(\log n)$ for both.

Approach today: search trees

Binary Search Tree Review

Binary search tree:

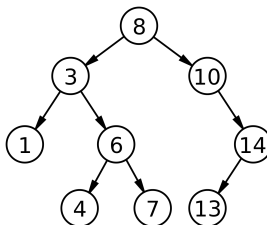
- ▶ All nodes have at most **2** children
- ▶ Each node stores (key, object) pair
- ▶ All descendants to left have smaller keys
- ▶ All descendants to the right have larger keys



Binary Search Tree Review

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Lookup: follow path from root!

Dictionary Operations in Simple Binary Search Tree

insert(x):

- ▶ If tree empty, put x at root
- ▶ Else if $x < \mathbf{root.key}$ recursively insert into left child
- ▶ Else (if $x > \mathbf{root.key}$) recursively insert into right child

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Example: H O P K I N S

Simple Binary Search Tree: Analysis

Pluses: easy to implement

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Want to make tree *balanced*.

Rest of today:

- ▶ B-trees: perfect balance, not binary
- ▶ Red-black trees: approximate balance, binary
- ▶ Turn out to be related!

B-Trees

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Definition (B-tree with parameter t)

1. Each node has between $t - 1$ and $2t - 1$ keys in it (except the root has between 1 and $2t - 1$ keys). Keys in a node are stored in a sorted array.
2. Each non-leaf has degree (number of children) equal to the number of keys in it plus 1 . If $\mathbf{v}$ is a node with keys $[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k]$ and the children are $[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k+1}]$, then the tree rooted at $\mathbf{v}_i$ contains only keys that are greater than $\mathbf{a}_{i-1}$ and less than $\mathbf{a}_i$ (except the edge cases: the tree rooted at $\mathbf{v}_1$ has keys less than $\mathbf{a}_1$, and the tree rooted at $\mathbf{v}_{k+1}$ has keys greater than $\mathbf{a}_k$).
3. All leaves are at the same depth.

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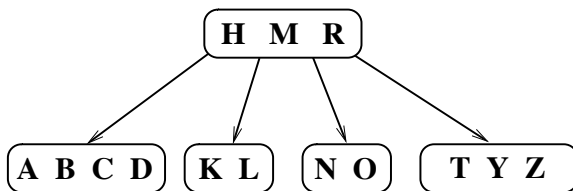
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When $t = 2$ known as a *2-3-4 tree*, since $\#$ children either 2, 3, or 4

B-tree: Example

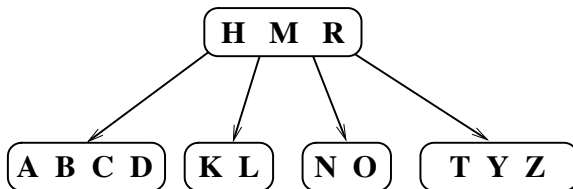
$t = 3$:

- ▶ Root has between **1** and **5** keys, non-roots have between **2** and **5** keys
- ▶ Non-leaves have between **3** and **6** children (root can have fewer).

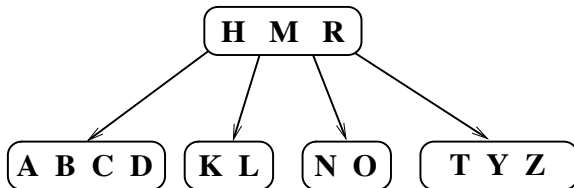


Lookups

Binary search in array at root. Finished if find item, else get pointer to appropriate child, recurse.



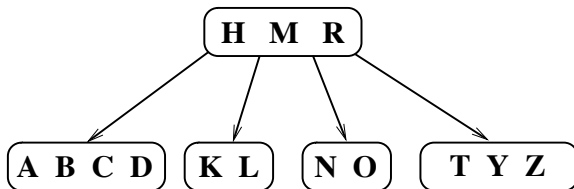
Insert(*x*)



Obvious approach: do a lookup, put *x* in leaf where it should be.

- ▶ Example: insert **E**

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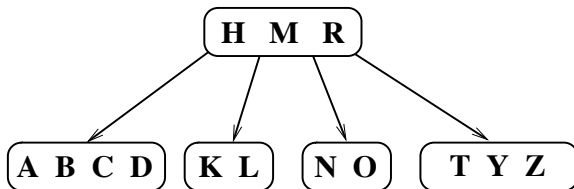


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Split:

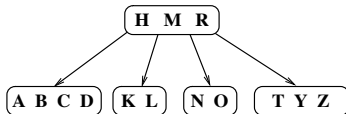
- ▶ Only used on *full* nodes (nodes with $2t - 1$ keys) whose parents are *not* full.
- ▶ Pull median of its keys up to its parent
- ▶ Split remaining $2t - 2$ keys into two nodes of $t - 1$ keys each. Reconnect appropriately.

Insert (continued)

Insert: do a lookup and insert at leaf, but when we encounter a full node on way down, split it.

Insert (continued)

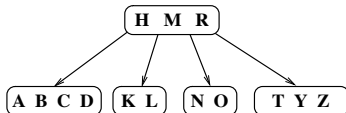
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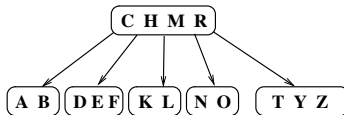
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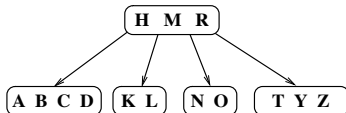


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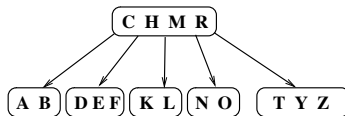


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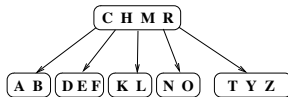


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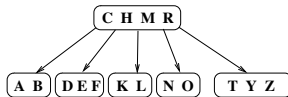


Note: since split *on the way down*, when a node is split, its parent is not full!

Example continued

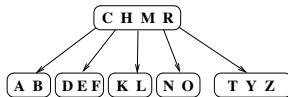


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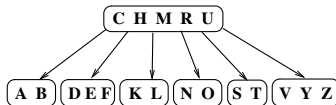


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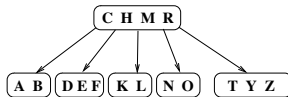
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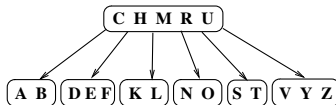
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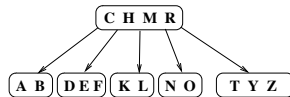


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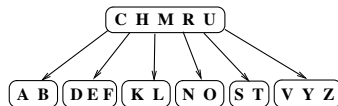


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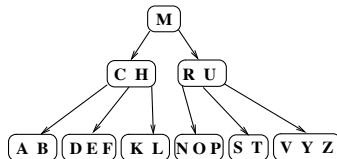
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Second property (correct degrees, subtrees have keys in correct ranges): Hooked nodes up correctly after split. ✓

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$t = 2$:

- ▶ 2-3-4 tree
- ▶ Can be implemented as *binary* tree using *red-black trees*

Red-Black Trees

Red-Black Trees: Intro

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Want *binary* balanced tree.

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- ▶ Classical and super important data structure question
- ▶ Many solutions!

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Most famous: *red-black trees*

- ▶ Default in Linux kernel, used to optimize Java HashMap, ...
- ▶ Today: Quick overview, connection to 2-3-4 trees.
- ▶ *Not* traditional or practical point of view on red-black trees. See book!

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Can we turn a 2-3-4 tree into a binary tree with all the same properties?

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- ▶ Degree 2: good!

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- ▶ Degree 4:

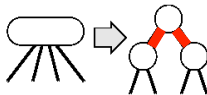
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Can we turn a 2-3-4 tree into a binary tree with all the same properties?

- ▶ *No*: can't have perfect balance!
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Nodes in 2-3-4 tree have degree 2, 3, or 4

- ▶ Degree 2: good!
- ▶ Degree 4:



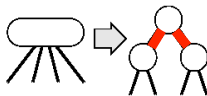
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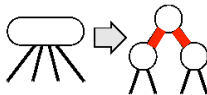
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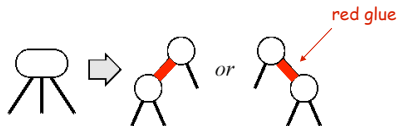
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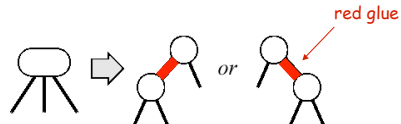
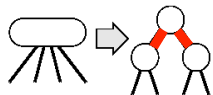
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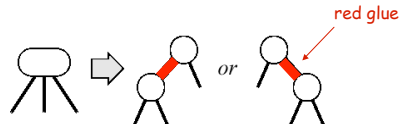
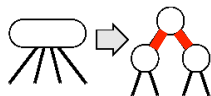
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Important Properties



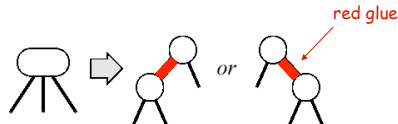
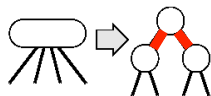
Important Properties



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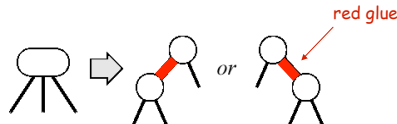
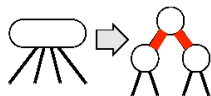
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2. Every leaf has same number of *black* edges on path to root (*black-depth*)
 - ▶ Each black edge is a 2-3-4 tree edge
 - ▶ All leaves in 2-3-4 tree at same distance from root

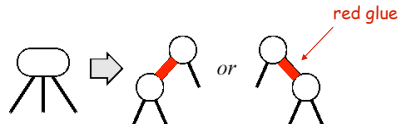
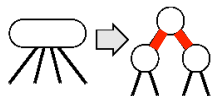
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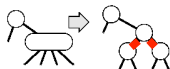
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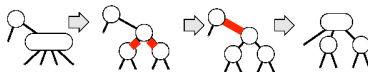


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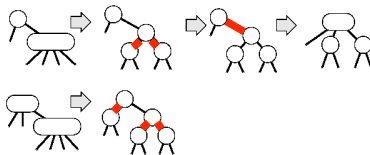


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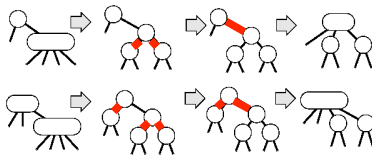


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Easy cases:

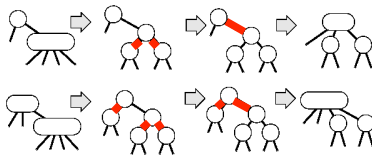


Insert

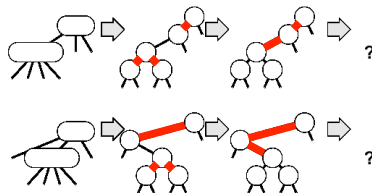
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2-3-4 trees: split full nodes on way down.

Easy cases:



Harder cases:

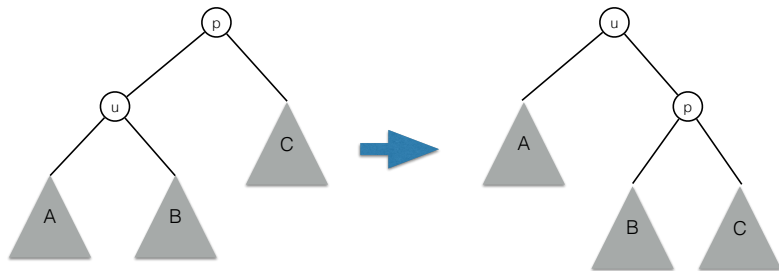


Tree Rotations

Used in many different tree constructions.

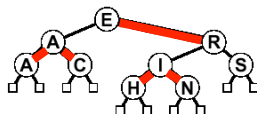
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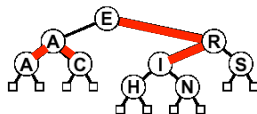
Using Rotations

Can use rotations to “fix” hard cases. Example:

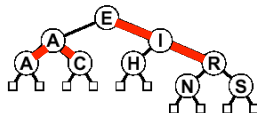


inserting *G*

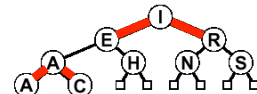
change colors



right rotate R →



left rotate E →



End

A few more complications to deal with – see lecture notes, textbook.

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Main points:

- ▶ Red-Black trees can be thought of as a binary implementation of 2-3-4 trees
- ▶ Approximately balanced, so $O(\log n)$ lookup time
- ▶ Insert time (basically) same as 2-3-4 tree, so also $O(\log n)$.
- ▶ See book for direct approach (not through 2-3-4 trees).